Capital Reserve Depletion and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Capital Reserve Depletion (CRD), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by Novy-Marx and Velikov (2023). A value-weighted long/short trading strategy based on CRD achieves an annualized gross (net) Sharpe ratio of 0.54 (0.47), and monthly average abnormal gross (net) return relative to the Fama and French (2015) five-factor model plus a momentum factor of 21 (20) bps/month with a t-statistic of 3.08 (2.99), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Share issuance (1 year), Growth in book equity, Share issuance (5 year), Net Payout Yield, Change in equity to assets, Net equity financing) is 16 bps/month with a t-statistic of 2.45.

1 Introduction

Asset prices reflect investors' marginal utility across states of nature, with risk premia emerging from the covariance between asset payoffs and consumption growth. While the consumption-based asset pricing paradigm provides the theoretical foundation for understanding cross-sectional return variation, empirical tests often struggle to connect firm characteristics directly to consumption risk exposure. We identify capital reserve depletion (CRD) as a novel predictor of stock returns that captures firms' vulnerability to aggregate consumption shocks. Firms experiencing rapid depletion of their capital reserves face heightened sensitivity to economic downturns, as their reduced financial buffers limit their ability to smooth cash flows during periods of low consumption growth. This financial fragility translates into higher required returns, as investors demand compensation for bearing consumption risk during states when their marginal utility is highest.

The economic mechanism linking capital reserve depletion to expected returns operates through firms' differential exposure to consumption risk. Following the long-run risks framework of Bansal and Yaron (2004), small persistent shocks to consumption growth generate substantial variation in asset prices through their effect on investors' intertemporal marginal rate of substitution. Firms depleting capital reserves exhibit greater sensitivity to these persistent consumption shocks, as their diminished financial flexibility amplifies the transmission of aggregate economic fluctuations to firm cash flows. This heightened exposure manifests particularly during economic contractions when the representative agent's marginal utility peaks, consistent with the habit formation models of Campbell and Cochrane (1999).

The state-dependent nature of capital reserve depletion risk aligns with recent evidence on time-varying disaster risk premia documented by Wachter (2013) and Gabaix (2012). During normal economic conditions, firms can access external financing to offset temporary cash flow shortfalls, but this substitutability breaks down precisely when consumption growth turns negative and disaster risk materializes. Capital-depleted firms face binding financing constraints exactly when the stochastic discount factor is highest, generating a positive risk premium through the covariance between their returns and marginal utility. The consumption-based factor models of Yogo and Piazzesi (2006) predict that assets with high loadings on consumption growth command higher expected returns, and our capital reserve depletion measure captures this exposure through firms' financial vulnerability.

The cross-sectional variation in CRD-related returns reflects heterogeneous exposure to the consumption-based pricing kernel rather than behavioral biases or market frictions. Following Parker and Julliard (2005), we interpret the return predictability of capital reserve depletion through the lens of ultimate consumption risk—the covariance between returns and long-horizon consumption growth. Firms experiencing rapid reserve depletion face higher bankruptcy risk during consumption disasters, leading to permanent capital losses for equity holders when marginal utility is highest. This interpretation connects our findings to the broader literature on intermediary asset pricing He and Krishnamurthy (2013), where financial constraints of intermediaries amplify the pricing of consumption risk. The depletion of capital reserves signals firms' reduced capacity to withstand adverse consumption shocks, requiring higher expected returns to compensate investors for bearing this systematic risk.

Our empirical analysis reveals that capital reserve depletion strongly predicts cross-sectional variation in expected returns. The value-weighted long-short strategy based on CRD achieves an annualized gross Sharpe ratio of 0.54, with a monthly average excess return of 29 basis points and a t-statistic of 4.23. The strategy's alpha relative to the Fama-French six-factor model equals 21 basis points per month with a t-statistic of 3.08, demonstrating that CRD captures a dimension of systematic risk not spanned by existing factors. The economic magnitude of these returns confirms that investors demand substantial compensation for bearing the consumption risk associated with capital-depleted firms.

The robustness of our findings across different portfolio construction methodologies strengthens the consumption-based interpretation of CRD's return predictability. Among the largest quintile of stocks by market capitalization, the CRD strategy generates an average return of 25 basis points per month with a t-statistic of 2.94, indicating that the phenomenon extends beyond small, illiquid securities where limits to arbitrage might explain pricing anomalies. After accounting for transaction costs using the composite bid-ask spreads of Chen and Velikov (2022), the net Sharpe ratio remains 0.47, with the strategy achieving positive and significant net generalized alphas relative to all five standard factor models. These results confirm that CRD-based strategies expand the mean-variance efficient frontier available to investors.

Controlling for the six most closely related anomalies and the Fama-French six factors simultaneously, the CRD strategy maintains an alpha of 16 basis points per month with a t-statistic of 2.45. This residual return spread demonstrates that capital reserve depletion contains unique information about firms' exposure to consumption risk beyond that captured by related measures of equity issuance and payout policy. The persistence of significant alphas after controlling for momentum, profitability, and investment factors suggests that CRD identifies a distinct source of systematic risk rooted in firms' vulnerability to aggregate consumption shocks rather than firm-specific characteristics unrelated to marginal utility.

Our study contributes to the consumption-based asset pricing literature by identifying capital reserve depletion as a direct measure of firms' exposure to consumption risk. While Lettau and Ludvigson (2001b) and Lettau and Ludvigson (2001a) establish the theoretical and empirical importance of consumption-based factors, our work provides a firm-level characteristic that captures heterogeneous loadings on these aggregate risk factors. Unlike the external financing anomalies documented in Daniel and Titman (2006) and Pontiff and Woodgate (2008), which focus on agency conflicts and market timing, we demonstrate that CRD's return predictability stems from its correlation with the intertemporal marginal rate of substitution. This distinction is crucial for understanding why certain firm characteristics predict returns in equilibrium models with rational, risk-averse investors.

Our methodological approach advances the literature by applying the comprehensive testing framework of Novy-Marx and Velikov (2023) to evaluate CRD's incremental contribution to the factor zoo. Previous studies such as McLean and Pontiff (2016) and Hou et al. (2020) emphasize the importance of out-of-sample validation and controlling for data mining concerns when proposing new return predictors. By demonstrating that CRD maintains significant alpha after controlling for 212 existing anomalies and survives multiple robustness tests including transaction costs, we establish its validity as a genuine risk factor rather than a statistical artifact. The finding that CRD ranks in the top 2% of net Sharpe ratios among all documented anomalies underscores its economic importance for portfolio allocation.

The broader implications of our findings extend to the debate over rational versus behavioral explanations for cross-sectional return patterns. The strong performance of CRD among large-cap stocks and its robustness to transaction costs support a risk-based interpretation consistent with Fama and French (2015) and Fama and French (2018), rather than mispricing that should be eliminated through arbitrage. Our results suggest that firm-level financial constraints create heterogeneous exposure to aggregate consumption risk, providing a micro-foundation for macro-finance models that link asset prices to consumption dynamics. This connection between capital structure decisions and systematic risk exposure offers new insights into how corporate financial policy affects required returns through its impact on firms' ability to withstand consumption disasters.

2 Data

We obtain accounting data from the COMPUSTAT Annual database, which provides comprehensive financial statement information for publicly traded U.S. companies. Our sample includes all firms with available data on common stock and capital surplus accounts, spanning several decades of financial reporting. We focus on non-financial firms and exclude observations with missing or negative values for the required accounting variables.Capital Reserve Depletion signal relies on two key balance sheet items. Common stock (CSTK) represents the par or stated value of all common stock issued by the firm, reflecting the nominal value assigned to shares at issuance. Capital surplus (CAPS), also known as additional paid-in capital, captures the amount shareholders have paid for stock in excess of its par value, representing the premium over the stated value received when shares are issued. Together, these components constitute important elements of shareholders' equity, with changes in their relationship potentially signaling shifts in a firm's equity financing activities or capital structure decisions.construct the Capital Reserve Depletion signal by computing the year-over-year change in common stock scaled by lagged capital surplus: (CSTK(t) - CSTK(t-1)) / CAPS(t-1). This ratio captures the rate at which firms alter their common stock base relative to their existing capital surplus cushion. A positive value indicates an increase in common stock relative to the prior year's capital surplus, while negative values suggest reductions through buybacks or retirements. We calculate this measure using fiscal year-end values to ensure consistency across firms with different fiscal year conventions. To maintain data quality, we require non-missing values for both CSTK and CAPS in consecutive years, and we exclude observations where the lagged capital surplus equals zero to avoid undefined ratios. We winsorize the signal at the 1st and 99th percentiles to mitigate the influence of extreme outliers.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the CRD signal. Panel A plots the time-series of the mean, median, and interquartile range for CRD. On average, the cross-sectional mean (median) CRD is -0.06 (-0.00) over the 1963 to 2024 sample, where the starting date is determined by the availability of the input CRD data. The signal's interquartile range spans -0.14 to 0.00. Panel B of Figure 1 plots the time-series of the coverage of the CRD signal for the CRSP universe. On average, the CRD signal is available for 5.78% of CRSP names, which on average make up 6.74% of total market capitalization.

4 Does CRD predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on CRD using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high CRD portfolio and sells the low CRD portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short CRD strategy earns an average return of 0.29% per month with a t-statistic of 4.23. The annualized Sharpe ratio of the strategy is 0.54. The alphas range from 0.21% to 0.32% per month and have t-statistics exceeding 3.08 everywhere. The lowest alpha is with respect to the FF6 factor model.

Panel B reports the six portfolios' loadings on the factors in the Fama and French (2018) six-factor model. The long/short strategy's most significant loading is 0.26, with a t-statistic of 5.64 on the CMA factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in $ millions) of the stocks they hold. In an average month, the five portfolios have at least 531 stocks and an average market capitalization of at least $1,363 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor's achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different protfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using cap breakpoints and value-weighted portfolios, and equals 25 bps/month with a t-statistics of 3.53. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-five exceed two, and for twenty-three exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of Novy-Marx and Velikov (2016). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from Chen and Velikov (2022). The net average returns reported in the first column range between 21-32bps/month. The lowest return, (21 bps/month), is achieved from the quintile sort using cap breakpoints and value-weighted portfolios, and has an associated t-statistic of 3.00. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the CRD trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty-five cases, and significantly expands the achievable frontier in twenty-five cases.

Table 3 provides direct tests for the role size plays in the CRD strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and CRD, as well as average returns and alphas for long/short trading CRD strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80^{th} NYSE percentile), the CRD strategy achieves an average return of 25 bps/month with a t-statistic of 2.94. Among these large cap stocks, the alphas for the CRD strategy relative to the five most common factor models range from 21 to 28 bps/month with t-statistics between 2.38 and 3.32.

5 How does CRD perform relative to the zoo?

Figure 2 puts the performance of CRD in context, showing the long/short strategy performance relative to other strategies in the "factor zoo." It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.[1] The vertical red line shows where the Sharpe ratio for the CRD strategy falls in the distribution. The CRD strategy's gross (net)

[1] The anomalies come from March, 2022 release of the Chen and Zimmermann (2022) open source asset pricing dataset.

Sharpe ratio of 0.54 (0.47) is greater than 92% (98%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a $1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a $1 invested in the CRD strategy (red line).[2] Ignoring trading costs, a $1 invested in the CRD strategy would have yielded $6.31 which ranks the CRD strategy in the top 4% across the 212 anomalies. Accounting for trading costs, a $1 invested in the CRD strategy would have yielded $4.49 which ranks the CRD strategy in the top 2% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and Novy-Marx and Velikov (2016) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the CRD relative to those. Panel A shows that the CRD strategy gross alphas fall between the 63 and 70 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 196306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The CRD strategy has a positive net generalized alpha for five out of the five factor models. In these cases CRD ranks between the 80 and 86 percentiles in terms of how much it could have expanded the achievable investment frontier.

[2]The figure assumes an initial investment of $1 in T-bills and $1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in Detzel et al. (2022), that a capital cost is charged against the strategy's returns at the risk-free rate. This excess return corresponds more closely to the strategy's economic profitability.

6 Does CRD add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of CRD with 210 filtered anomaly signals.[3] Figure 6 also shows an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price CRD or at least to weaken the power CRD has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of CRD conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{CRD} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{CRD} CRD_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{CRD,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on CRD. Stocks are finally grouped into five CRD portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted

[3]When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

CRD trading strategies conditioned on each of the 210 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on CRD and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the CRD signal in these Fama-MacBeth regressions exceed -0.05, with the minimum t-statistic occurring when controlling for Share issuance (5 year). Controlling for all six closely related anomalies, the t-statistic on CRD is 0.24.

Similarly, Table 5 reports results from spanning tests that regress returns to the CRD strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the CRD strategy earns alphas that range from 13-24bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 2.02, which is achieved when controlling for Share issuance (5 year). Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the CRD trading strategy achieves an alpha of 16bps/month with a t-statistic of 2.45.

7 Does CRD add relative to the whole zoo?

Finally, we can ask how much adding CRD to the entire factor zoo could improve investment performance. Figure 8 plots the growth of $1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). The combinations use either the 154 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 154 anomalies augmented with the CRD signal.[4] We consider one different methods for combining signals.

[4]We filter the 207 Chen and Zimmermann (2022) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which CRD is available.

Panel A shows results using "Average rank" as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, $1 investment in the 154-anomaly combination strategy grows to $2560.30, while $1 investment in the combination strategy that includes CRD grows to $2389.63.

8 Conclusion

This study provides compelling evidence that Capital Reserve Depletion (CRD) represents a economically meaningful and statistically robust predictor of cross-sectional equity returns. Our analysis, conducted using the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that CRD-based trading strategies generate substantial risk-adjusted returns that persist even after accounting for well-established risk factors and related anomalies.

The key findings reveal that a value-weighted long/short portfolio sorted on CRD delivers an annualized Sharpe ratio of 0.54 gross (0.47 net of transaction costs), indicating strong economic significance that survives real-world implementation frictions. More importantly, the strategy generates statistically significant abnormal returns of 21 basis points per month (t-statistic = 3.08) relative to the Fama-French five-factor model augmented with momentum. The robustness of these results is further confirmed by the persistence of a 16 basis point monthly alpha (t-statistic = 2.45) even after controlling for six closely related factors from the factor zoo, including various measures of equity issuance and financing activities.

These findings have important practical implications for both academic research and investment practice. For researchers, CRD emerges as a distinct source of cross-sectional return predictability that cannot be fully explained by existing factor models or related corporate financing signals. For practitioners, the strategy's net Sharpe ratio of 0.47 suggests that CRD can be profitably implemented even after accounting for realistic transaction costs, offering a viable addition to quantitative investment strategies.

However, several limitations warrant consideration. First, while our analysis accounts for transaction costs, we do not model potential market impact or capacity constraints that might affect the scalability of CRD-based strategies in practice. Second, the economic mechanisms underlying the CRD anomaly require further theoretical development to understand whether this represents a risk-based explanation or a behavioral mispricing that might attenuate over time.

Future research should explore several promising avenues. Investigating the time-varying nature of the CRD premium and its relationship to macroeconomic conditions could provide insights into the underlying economic drivers. Additionally, examining the international evidence for CRD would help establish whether this phenomenon extends beyond U.S. equity markets. Finally, developing a theoretical framework that explains why capital reserve depletion predicts future returns would enhance our understanding of this anomaly and its place within the broader asset pricing literature. Such investigations would not only validate the robustness of our findings but also contribute to the ongoing debate about the sources of cross-sectional return predictability in equity markets.

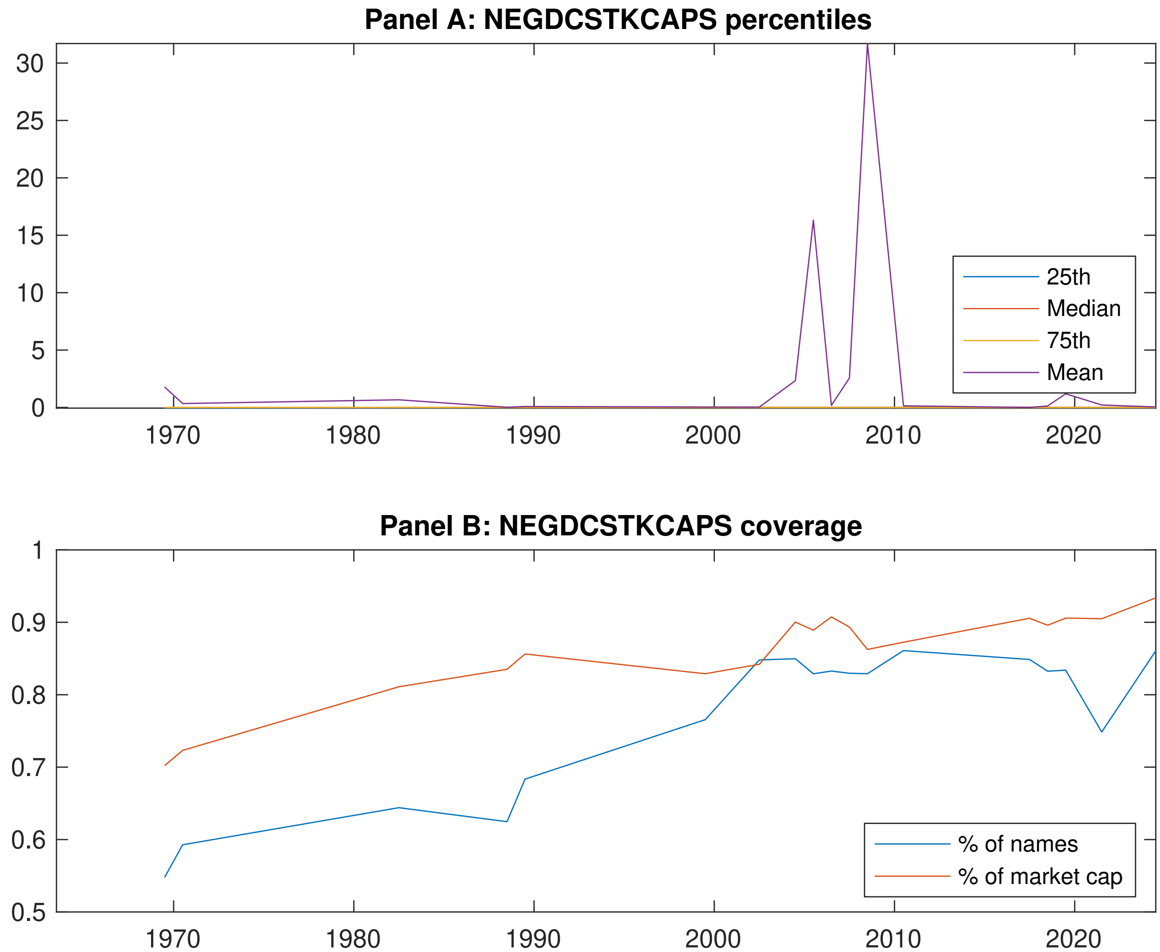


Figure 1: Times series of CRD percentiles and coverage.
This figure plots descriptive statistics for CRD. Panel A shows cross-sectional percentiles of CRD over the sample. Panel B plots the monthly coverage of CRD relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on CRD. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Excess returns and alphas on CRD-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.44	0.49	0.62	0.70	0.73	0.29
	[2.59]	[2.65]	[3.35]	[4.25]	[4.50]	[4.23]
α_{CAPM}	-0.14	-0.14	-0.01	0.14	0.18	0.32
	[-3.13]	[-2.75]	[-0.13]	[2.84]	[3.69]	[4.78]
α_{FF3}	-0.17	-0.13	0.02	0.11	0.14	0.31
	[-3.83]	[-2.41]	[0.39]	[2.29]	[3.03]	[4.63]
α_{FF4}	-0.13	-0.07	0.03	0.08	0.14	0.27
	[-2.86]	[-1.28]	[0.49]	[1.57]	[2.91]	[3.90]
α_{FF5}	-0.18	-0.04	0.05	0.07	0.06	0.24
	[-3.87]	[-0.85]	[0.96]	[1.44]	[1.39]	[3.57]
α_{FF6}	-0.14	-0.00	0.06	0.05	0.07	0.21
	[-3.08]	[-0.00]	[1.01]	[0.91]	[1.48]	[3.08]
Panel B: Fama and French (2018) 6-factor model loadings for CRD-sorted portfolios						
β_{MKT}	1.01	1.03	1.03	1.00	0.99	-0.01
	[90.64]	[83.35]	[75.17]	[83.95]	[88.92]	[-0.69]
β_{SMB}	-0.01	-0.01	0.07	-0.06	-0.04	-0.03
	[-0.53]	[-0.54]	[3.69]	[-3.43]	[-2.19]	[-1.13]
β_{HML}	0.10	-0.01	-0.12	0.09	0.03	-0.07
	[4.50]	[-0.42]	[-4.69]	[4.03]	[1.28]	[-2.16]
β_{RMW}	0.07	-0.14	-0.11	0.07	0.11	0.05
	[3.05]	[-5.70]	[-4.14]	[2.83]	[5.10]	[1.40]
β_{CMA}	-0.06	-0.14	0.04	0.06	0.20	0.26
	[-1.99]	[-3.95]	[1.08]	[1.69]	[6.34]	[5.64]
β_{UMD}	-0.05	-0.06	-0.01	0.04	-0.01	0.05
	[-4.77]	[-5.07]	[-0.40]	[3.14]	[-0.66]	[2.76]
Panel C: Average number of firms (n) and market capitalization (me)						
n	531	646	616	638	676	
me ($\$10^6$)	1475	1363	1857	2170	2324	

Table 2: Robustness to sorting methodology & trading costs
This table evaluates the robustness of the choices made in the CRD strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.29	0.32	0.31	0.27	0.24	0.21
			[4.23]	[4.78]	[4.63]	[3.90]	[3.57]	[3.08]
Quintile	NYSE	EW	0.48	0.49	0.46	0.40	0.41	0.37
			[7.64]	[7.86]	[7.66]	[6.65]	[6.87]	[6.13]
Quintile	Name	VW	0.26	0.30	0.28	0.25	0.22	0.20
			[3.82]	[4.42]	[4.18]	[3.61]	[3.20]	[2.84]
Quintile	Cap	VW	0.25	0.28	0.29	0.23	0.25	0.21
			[3.53]	[3.97]	[3.96]	[3.12]	[3.47]	[2.84]
Decile	NYSE	VW	0.36	0.38	0.37	0.32	0.36	0.32
			[4.24]	[4.49]	[4.29]	[3.64]	[4.14]	[3.64]

Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r^e_{net}	α^*_{CAPM}	α^*_{FF3}	α^*_{FF4}	α^*_{FF5}	α^*_{FF6}
Quintile	NYSE	VW	0.25	0.28	0.27	0.24	0.22	0.20
			[3.65]	[4.24]	[4.04]	[3.64]	[3.26]	[2.99]
Quintile	NYSE	EW	0.29	0.32	0.28	0.25	0.23	0.21
			[4.27]	[4.68]	[4.28]	[3.87]	[3.52]	[3.23]
Quintile	Name	VW	0.22	0.26	0.24	0.22	0.19	0.18
			[3.25]	[3.89]	[3.60]	[3.29]	[2.88]	[2.69]
Quintile	Cap	VW	0.21	0.25	0.24	0.21	0.22	0.20
			[3.00]	[3.48]	[3.42]	[2.96]	[3.13]	[2.79]
Decile	NYSE	VW	0.32	0.35	0.33	0.30	0.33	0.30
			[3.73]	[4.09]	[3.89]	[3.55]	[3.82]	[3.56]

Table 3: Conditional sort on size and CRD

This table presents results for conditional double sorts on size and CRD. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on CRD. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high CRD and short stocks with low CRD .Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 196306 to 202406.

Panel A: portfolio average returns and time-series regression results

		CRD Quintiles					CRD Strategies					
		(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Size quintiles	(1)	0.55	0.68	0.80	0.87	0.99	0.47	0.48	0.46	0.40	0.40	0.35
		[2.33]	[2.54]	[3.00]	[3.45]	[4.35]	[6.48]	[6.60]	[6.28]	[5.46]	[5.30]	[4.70]
	(2)	0.70	0.62	0.85	0.92	0.95	0.25	0.27	0.24	0.17	0.23	0.17
		[3.18]	[2.57]	[3.51]	[3.91]	[4.39]	[3.03]	[3.21]	[2.80]	[1.97]	[2.71]	[2.04]
	(3)	0.60	0.59	0.71	0.92	0.87	0.27	0.29	0.27	0.23	0.25	0.22
		[2.99]	[2.69]	[3.15]	[4.27]	[4.45]	[3.77]	[4.00]	[3.77]	[3.13]	[3.36]	[2.87]
	(4)	0.53	0.61	0.73	0.84	0.78	0.25	0.27	0.23	0.23	0.12	0.14
		[2.82]	[3.05]	[3.52]	[4.34]	[4.23]	[3.47]	[3.71]	[3.22]	[3.18]	[1.71]	[1.86]
	(5)	0.42	0.52	0.46	0.59	0.67	0.25	0.28	0.28	0.21	0.26	0.21
		[2.46]	[2.97]	[2.48]	[3.47]	[4.19]	[2.94]	[3.28]	[3.32]	[2.48]	[3.04]	[2.38]

Panel B: Portfolio average number of firms and market capitalization

		CRD Quintiles: Average n					CRD Quintiles: Average market capitalization ($\$10^6$)				
		(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)
Size quintiles	(1)	350	348	347	346	341	31	31	35	27	26
	(2)	100	97	99	98	97	53	52	53	52	51
	(3)	71	71	70	71	70	90	89	89	93	91
	(4)	59	59	59	59	59	188	189	198	197	201
	(5)	53	53	53	53	53	1213	1365	1479	1509	1793

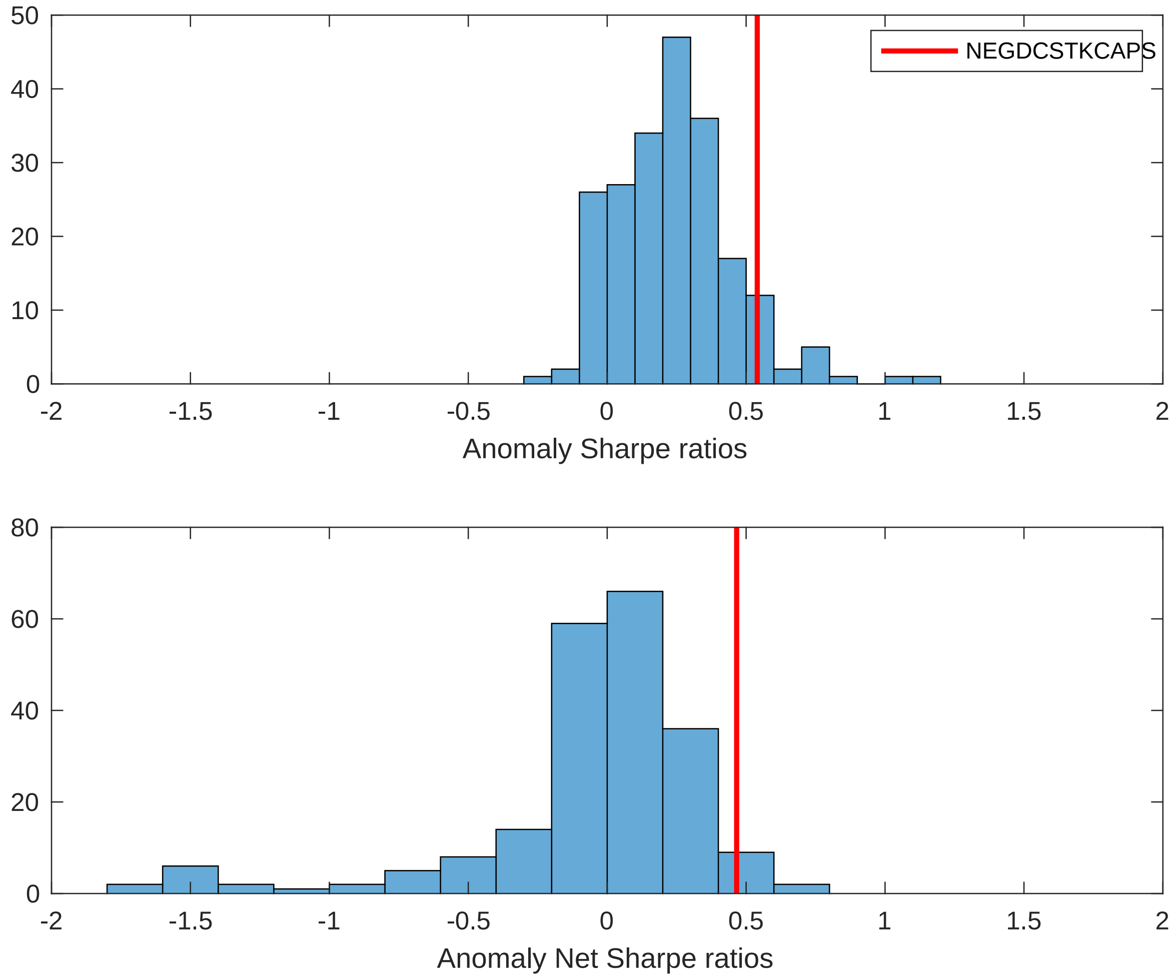


Figure 2: Distribution of Sharpe ratios.
This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the CRD with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

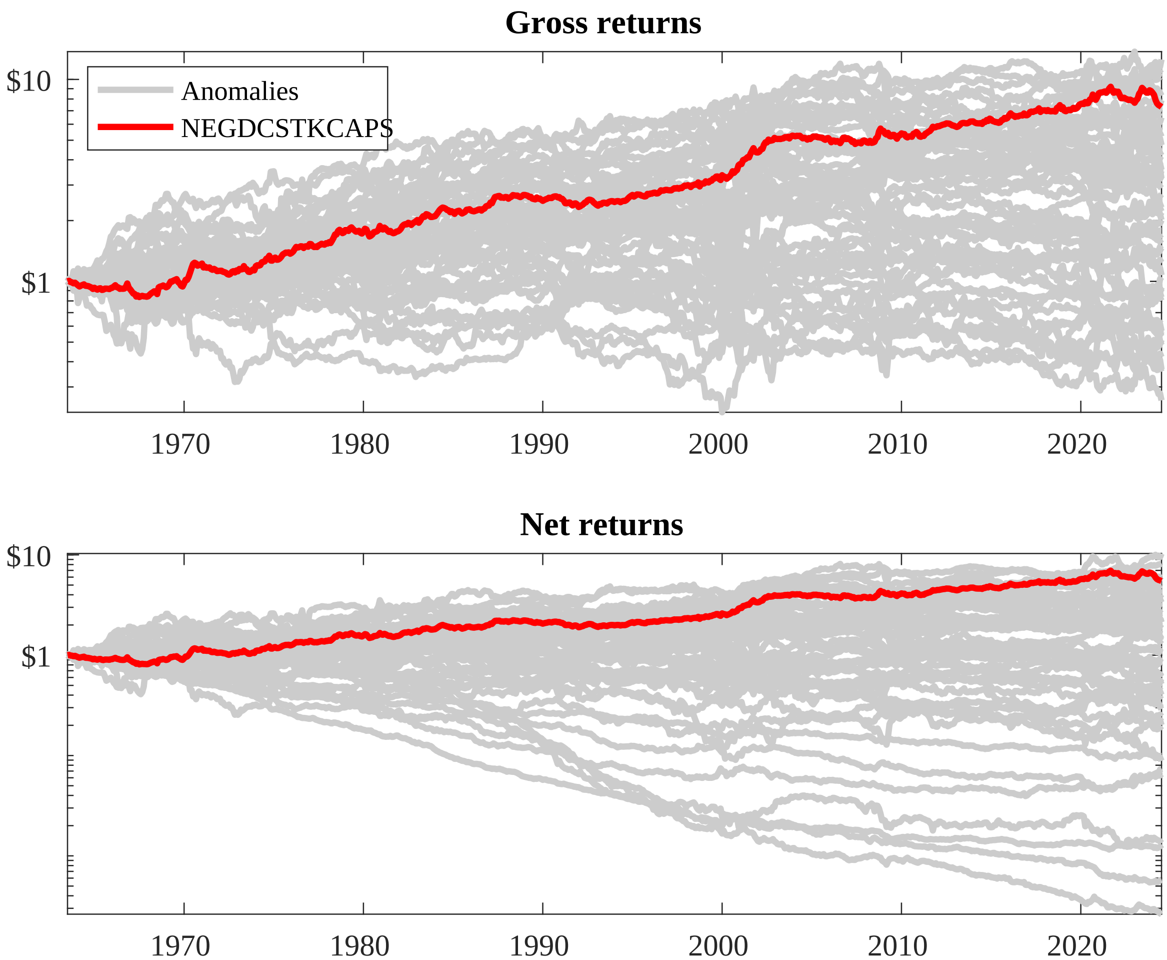
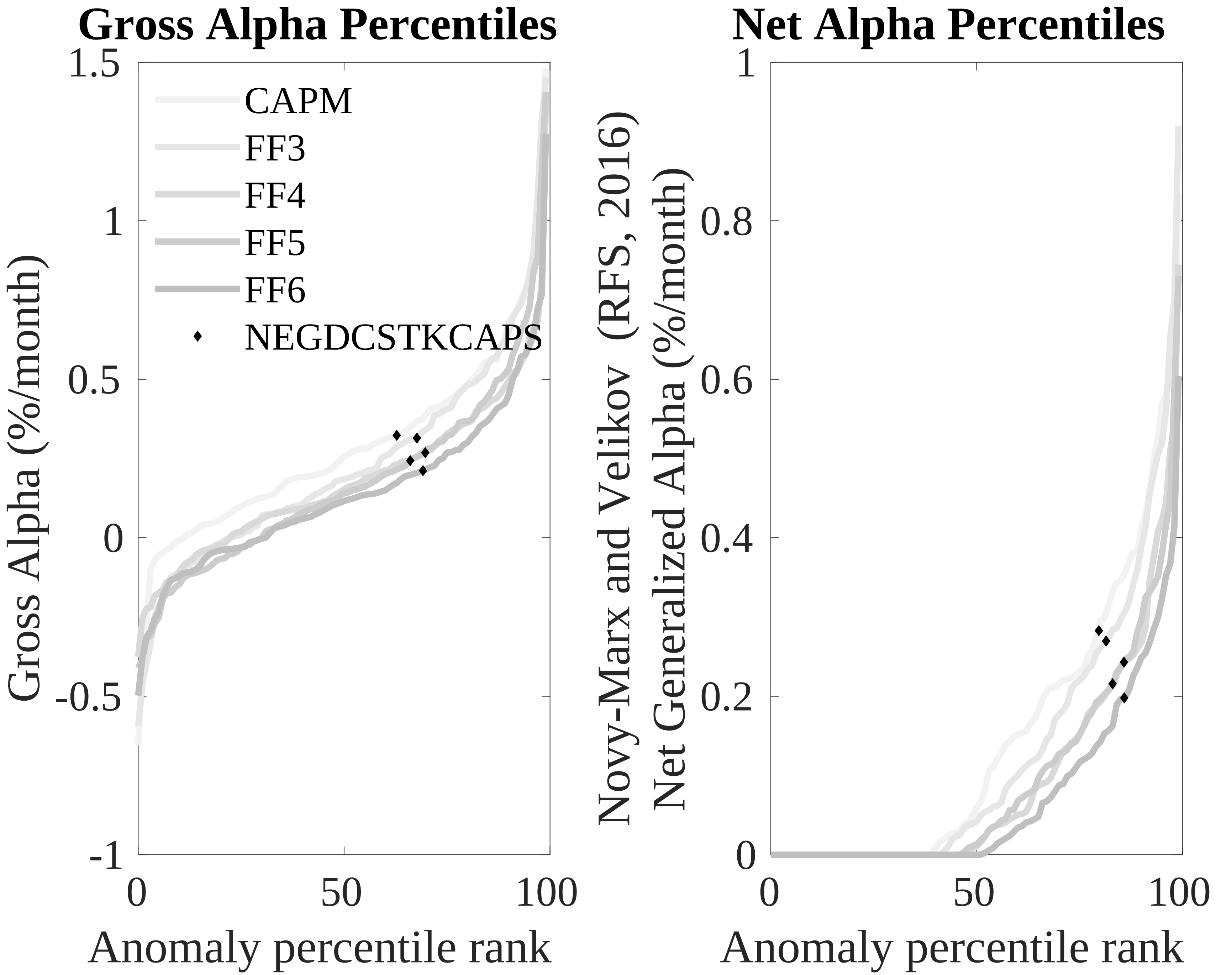


Figure 3: Dollar invested.
This figure plots the growth of a $1 invested in 212 anomaly trading strategies (gray lines), and compares those with the CRD trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strtaegy returns.


Gross Alpha Percentiles
Net Alpha Percentiles
CAPM
FF3
FF4
FF5
FF6
NEGDCSTKCAPS
Gross Alpha (%/month)
Novy-Marx and Velikov (RFS, 2016)
Net Generalized Alpha (%/month)
Anomaly percentile rank
Anomaly percentile rank
1.5
1
0.5
0
-0.5
-1
0.8
0.6
0.4
0.2
0
50
100

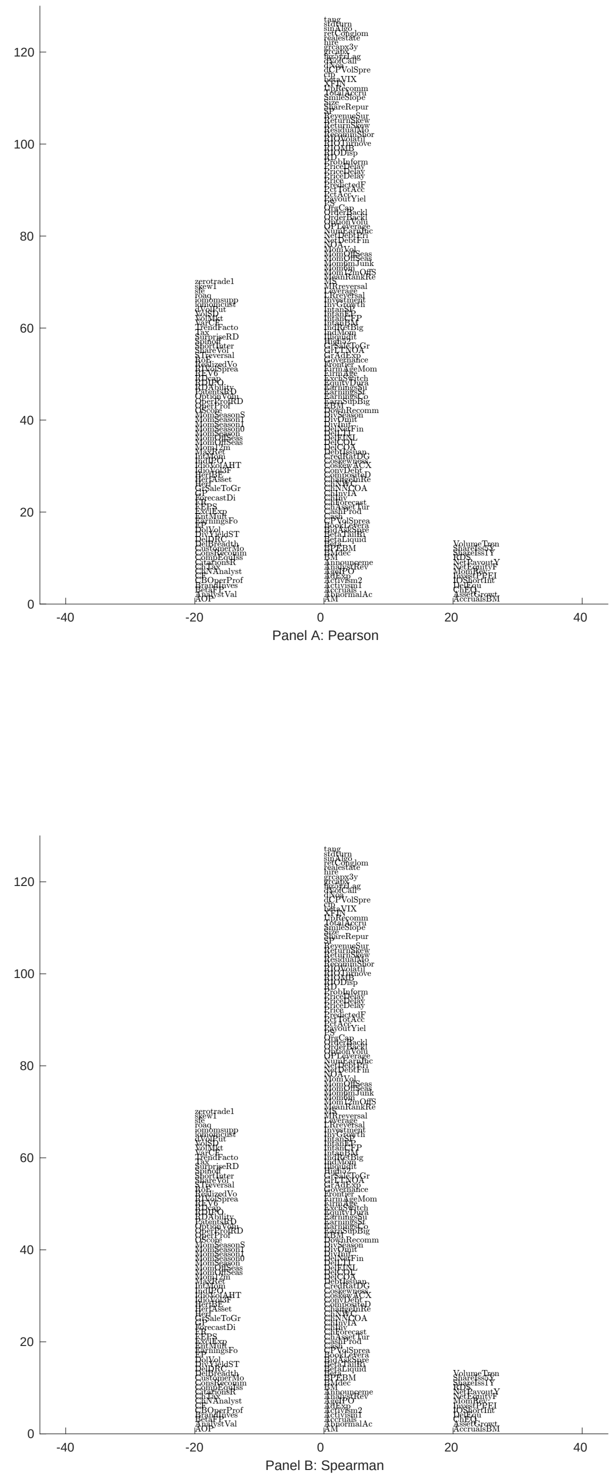


Figure 5: Distribution of correlations.
This figure plots a name histogram of correlations of 210 filtered anomaly signals with CRD. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

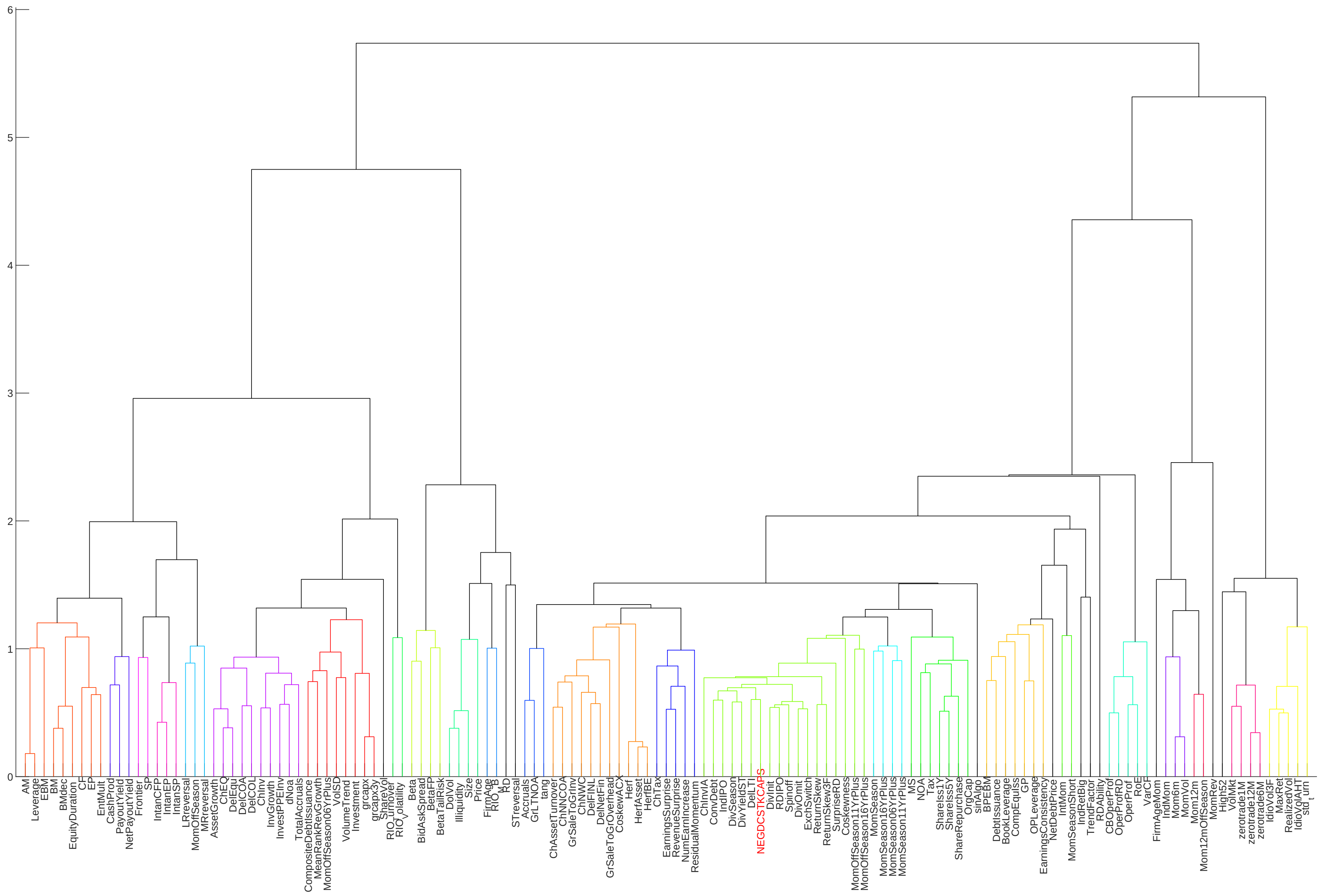


Figure 6: Agglomerative hierarchical cluster plot

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

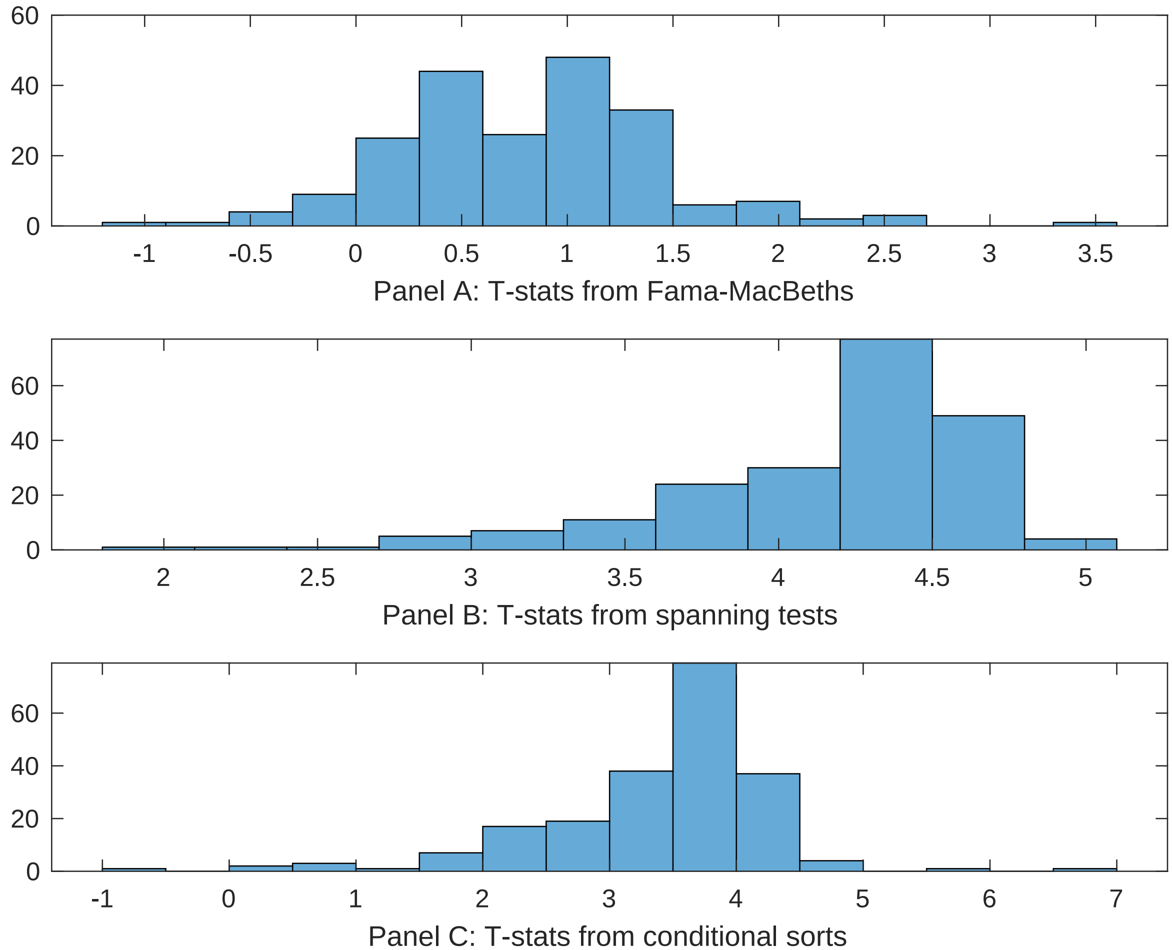


Figure 7: Distribution of t-stats on conditioning strategies
This figure plots histograms of t-statistics for predictability tests of CRD conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{CRD} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{CRD}CRD_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{CRD,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on CRD. Stocks are finally grouped into five CRD portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted CRD trading strategies conditioned on each of the 210 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on CRD. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{CRD} CRD_{i,t} + \sum_{k=1}^{s} ix \beta_{X_k} X_{i,t}^{k} + \epsilon_{i,t}$. The six most closely related anomalies, X, are Share issuance (1 year), Growth in book equity, Share issuance (5 year), Net Payout Yield, Change in equity to assets, Net equity financing. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.13	0.17	0.13	0.12	0.12	0.13	0.12
	[5.89]	[6.85]	[6.30]	[5.19]	[5.53]	[5.27]	[4.66]
CRD	0.19	0.99	-0.39	0.38	0.35	0.62	0.22
	[0.27]	[0.13]	[-0.05]	[0.41]	[0.44]	[0.68]	[0.24]
Anomaly 1	0.15						0.79
	[6.29]						[3.52]
Anomaly 2		0.43					-0.13
		[3.52]					[-0.91]
Anomaly 3			0.28				0.27
			[5.12]				[5.93]
Anomaly 4				0.23			0.13
				[2.02]			[2.68]
Anomaly 5					0.12		0.13
					[3.36]		[2.21]
Anomaly 6						0.18	-0.13
						[3.06]	[-1.61]
# months	715	720	715	704	720	630	622
$\bar{R}^2$(%)	0	0	0	1	0	1	0

Table 5: Spanning tests controlling for most closely related anomalies
This table presents spanning tests results of regressing returns to the CRD trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{CRD} = \alpha + \sum_{k=1}^{6} \beta_{X_k} r_t^{X_k} + \sum_{j=1}^{6} \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor. The six most closely related anomalies, X, are Share issuance (1 year), Growth in book equity, Share issuance (5 year), Net Payout Yield, Change in equity to assets, Net equity financing. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.19	0.20	0.13	0.24	0.21	0.22	0.16
	[3.00]	[3.00]	[2.02]	[3.61]	[3.08]	[3.08]	[2.45]
Anomaly 1	32.53						15.11
	[10.10]						[3.45]
Anomaly 2		21.20					20.38
		[5.90]					[3.82]
Anomaly 3			31.99				21.63
			[9.32]				[5.00]
Anomaly 4				15.75			12.14
				[6.14]			[3.63]
Anomaly 5					6.82		-11.47
					[2.03]		[-2.34]
Anomaly 6						11.19	-12.60
						[3.45]	[-3.21]
mkt	-0.15	-0.23	0.49	1.51	-1.05	1.34	1.48
	[-0.10]	[-0.14]	[0.32]	[0.93]	[-0.65]	[0.76]	[0.89]
smb	-0.29	-2.25	0.56	0.93	-1.63	6.92	1.01
	[-0.13]	[-0.98]	[0.25]	[0.40]	[-0.70]	[2.47]	[0.38]
hml	-6.79	-8.39	-3.35	-11.95	-6.68	-7.99	-9.89
	[-2.32]	[-2.72]	[-1.13]	[-3.70]	[-2.12]	[-2.49]	[-3.12]
rmw	-13.04	5.60	-6.88	-4.57	5.22	3.30	-7.67
	[-3.79]	[1.81]	[-2.13]	[-1.33]	[1.65]	[0.87]	[-1.91]
cma	5.61	5.26	10.29	12.46	18.93	13.89	-8.01
	[1.19]	[0.92]	[2.22]	[2.52]	[3.32]	[2.67]	[-1.35]
umd	3.13	4.27	2.53	5.88	4.82	4.37	2.98
	[2.08]	[2.72]	[1.65]	[3.73]	[3.01]	[2.64]	[1.91]
# months	728	732	728	728	732	630	626
$\bar{R}^2(\%)$	19	12	18	12	8	8	22

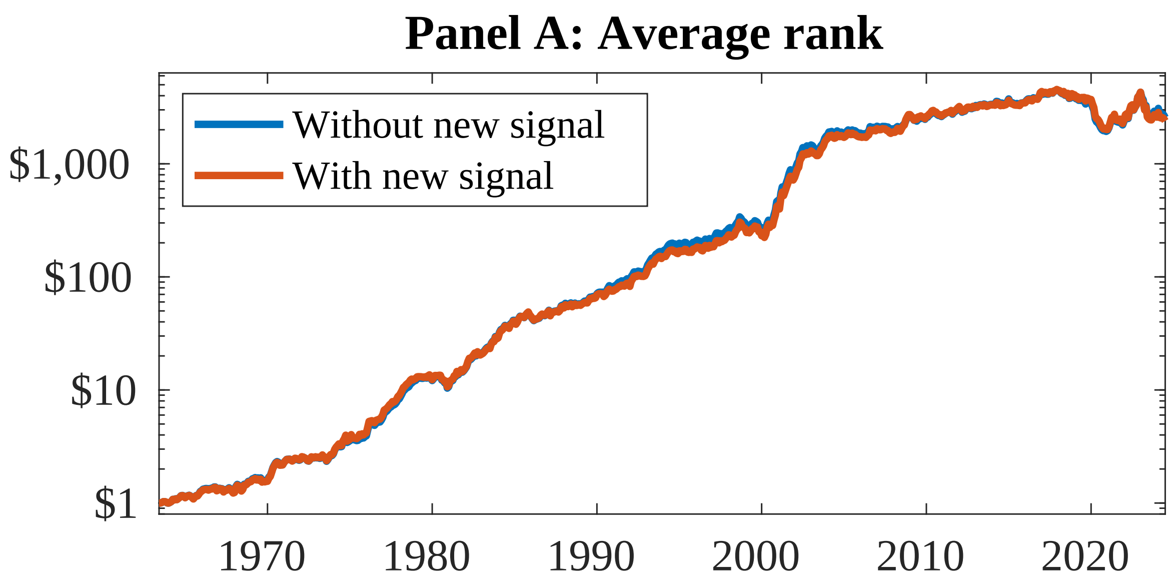


Figure 8: Combination strategy performance
This figure plots the growth of a $1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). In all panels, the blue solid lines indicate combination trading strategies that utilize 154 anomalies. The red solid lines indicate combination trading strategies that utilize the 154 anomalies as well as CRD. Panel A shows results using "Average rank" as the combination method. See Section 7 for details on the combination methods.

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